

1. (1) $\max z = 2x_1 + 3x_2 - 5x_3 - Mx_4 - Mx_5$

$$\begin{cases} x_1 + x_2 + x_3 + x_4 = 7 \\ 2x_1 - 5x_2 + x_3 + x_5 = 10 \\ x_1, x_2, x_3, x_4, x_5 \geq 0 \end{cases}$$

$$\begin{array}{cccccccc} & 2 & 3 & -5 & -M & -M & & \\ x_4 & -M & 7 & 1 & 1 & 1 & 0 & 7 \\ x_5 & -M & 10 & 2 & -5 & 1 & 0 & 10 \\ & 2+3M & 3-M & -5+2M & 0 & 0 & & \end{array}$$

x_1 换入, x_5 换出

$$\begin{array}{cccccccc} & 2 & 3 & -5 & -M & -M & & \\ x_4 & -M & 2 & 0 & \frac{1}{2} & \frac{1}{2} & 1 & -\frac{1}{2} \\ x_1 & 2 & 5 & 1 & -\frac{5}{2} & \frac{1}{2} & 0 & \frac{1}{2} \\ & 0 & 8+3M & -6+\frac{M}{2} & 0 & -\frac{M}{2}-1 & & \end{array}$$

x_2 换入, x_4 换出

$$\begin{array}{cccccccc} & 2 & 3 & -5 & -M & -M & & \\ x_2 & 3 & \frac{4}{7} & 0 & 1 & \frac{1}{2} & \frac{2}{7} & -\frac{1}{7} \\ x_1 & 2 & \frac{45}{7} & 1 & 0 & \frac{6}{7} & \frac{5}{7} & \frac{1}{7} \\ & 0 & 0 & -\frac{50}{7} & -M & \frac{16}{7} & -M+\frac{1}{7} & \end{array}$$

$$\therefore z_{\max} = 2 \times \frac{45}{7} + 3 \times \frac{4}{7} = \frac{102}{7}$$

2. $A = \begin{pmatrix} 2 & -1 & 2 \\ 1 & 0 & 4 \end{pmatrix} \quad B^T A = \begin{pmatrix} -1 & 2 \\ 0 & 1 \end{pmatrix} A = \begin{pmatrix} 0 & 1 & 6 \\ 1 & 0 & 4 \end{pmatrix}$

$$\begin{array}{cccc} & 6 & -2 & 3 \\ x_2 & -2 & 6 & 0 & 1 & 6 \\ x_1 & 6 & 4 & 1 & 0 & 4 \\ & 0 & 0 & -9 & & \end{array}$$

$$\therefore z_{\max} = 6 \times 4 - 2 \times 6 = 12$$

3. a) $\vec{p}$ 为二维向量

b) $\vec{c}^T - \vec{p}^T A \leq 0$

c) $(1 \ 3 \ 0 \ 0) - (p_1 \ p_2) \begin{pmatrix} 1 & 1 & -1 & 0 \\ 1 & 2 & 0 & 1 \end{pmatrix} \leq 0$

$$\begin{cases} 1 - p_1 - p_2 \leq 0 \\ 3 - p_1 - 2p_2 \leq 0 \\ p_1, p_2 \geq 0 \end{cases}$$

d) 要求 $\vec{p} \geq 0$

e) $g(p) = \vec{p}^T \vec{b} = (p_1 \ p_2) \begin{pmatrix} 8 \\ 12 \end{pmatrix} = 8p_1 + 12p_2$

f) 对偶问题为:

$$\min g(p) = 8p_1 + 12p_2$$

$$\text{s.t.} \begin{cases} p_1 + p_2 \geq 1 \\ p_1 + 2p_2 \geq 3 \\ p_1, p_2 \geq 0 \end{cases}$$

4. (1) $\min w = 8y_1 + 12y_2$

(2) $\min w = 8y_1 + 12y_2$

$$\text{s.t.} \begin{cases} y_1 + y_2 \geq 1 \\ y_1 + 2y_2 \geq 3 \\ y_1, y_2 \geq 0 \end{cases}$$

$$\text{s.t.} \begin{cases} y_1 + y_2 \geq 1 \\ y_1 + 2y_2 \geq 3 \\ y_1 \geq 0 \\ y_2 \leq 0 \end{cases}$$